

Nathan Jacobson

COMPUTER SCIENCE STUDENT · DEVELOPER

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Experience

Tae Kwon Do Instructor

Port Moody, BC

TRANSCENDING MARTIAL ARTS

September 2025 - Present

- Instructed children aged 4-14 martial arts techniques such as strikes, blocks, patterns, and weapons training
- Lead a group of assistant instructors to aid with instruction and managing the class

Information Officer

Port Moody, BC

ELECTIONS CANADA

April 2025

- Confirmed voter identification
- Directed voter traffic
- Assisted with counting votes
- Assisted voters with physical disabilities

Projects

intellidoc

🌐 grqphical/intellidoc

PYTHON, FASTAPI, SQLITE, LANCEDB

November 2025 - Present

- Currently developing a semantic search web application using Google's EmbeddingGemma model with both vector and keyword retrieval strategies, employing recursive chunking and metadata filtering to maximize accuracy.

interchange

🌐 grqphical/interchange

Go

August 2025-Present

- Created a reverse proxy in Golang that supports proxying, static file hosting, HTTPS, and IP blacklisting/whitelisting
- Currently working on adding rate limiting and load balancing

Remote Control and Autonomous Robotics System

🌐 grqphical/VRC-Code-2023

C++

September 2022 - February 2023

- Programmed both the autonomous and remote control software for my VEX VRC robotics team
- Robot was programmed to push discs into goal zones around the arena as well as roll a wall mounted roller to our team's colour
- Collaborated with the build team to troubleshoot issues with sensors
- Team placed 42nd in the province of British Columbia in skills competitions

Skills

Programming Languages

C, Python, Go, HTML, CSS, JS, SQL

Programs

Vim, Jupyter Notebooks, Excel, Word, Powerpoint

Other Skills

1st Degree Tae Kwon Do Black Belt, Photoshop, Video Editing, Computer Troubleshooting

Education

Simon Fraser University

Burnaby, BC

BSC IN COMPUTING SCIENCE

September 2025 - Present

- Recieved Early Admission for Fall 2025

Heritage Woods Secondary

Port Moody, BC

BC DOGWOOD DIPLOMA (HIGH SCHOOL DIPLOMA)

September 2021 - June 2025

- Awarded Top Producing Grade 12 Student
- Graduated with Honour Roll with Distinction
- Co-President of the Model United Nations Club
- Graduated with a 93% average

